

# A Reliable Schmitt-Trigger-Assisted CMOS SRAM PUF Cell with Lightweight Unstable-Bit Detection for IoT Hardware Security

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**Abstract**—Physically unclonable functions (PUFs) let a low-cost integrated circuit derive a chip-unique key from random manufacturing variation, avoiding stored secrets in insecure non-volatile memory. A known drawback of static random-access memory (SRAM) PUFs is that a fraction of the bits are unstable across power-ups and reads. This work presents a compact CMOS SRAM PUF and studies whether a Schmitt-trigger-assisted readout together with a lightweight unstable-bit detector improves reliability. A conventional 6T SRAM cell is used as the baseline; the proposed cell adds a Schmitt-trigger buffer on the readout path and a dual-latch, XOR-based detector that flags bits that disagree between two reads. A 4×4 array of both designs was captured, laid out, and simulated in the AMIS C5 0.5  $\mu\text{m}$  process ( $V_{DD} = 5\text{ V}$ ). Across 60 Monte-Carlo instances the array shows near-ideal uniqueness (49.8 %) and uniformity (45.9 %). Under a 5 mV read-noise assumption the raw bit-error rate is 15.0 %; after the detector masks the unstable bits, the retained key reaches 1.5 % BER (98.5 % reliability) on an independent set of reads. The proposed cell costs +31 % area (1-bit) and a slower, gated readout, and the results confirm that the reliability gain is produced by the detector rather than by the Schmitt trigger itself. All electrical results are schematic-level SPICE simulations; area is measured from the drawn layout.

**Index Terms**—Physically unclonable function, SRAM PUF, Schmitt trigger, unstable-bit detection, hardware security, IoT, bit error rate, CMOS.

## I. INTRODUCTION

Connected devices in the Internet of Things (IoT) increasingly need a cryptographic identity to authenticate themselves and to protect their communication. The classical approach stores a secret key in on-chip non-volatile memory, but such storage adds cost and remains readable by an attacker who can open the package. A physically unclonable function offers a different route: instead of storing a key, the circuit generates one on demand from the tiny, random mismatch that every fabrication process introduces between nominally identical transistors. Because the mismatch is set by uncontrollable physical variation, the resulting response is unique to each chip and difficult to clone, even by the manufacturer.

SRAM-based PUFs are attractive for low-cost nodes because they reuse a standard memory bit-cell. When a cross-coupled 6T cell powers up from an undefined state, its two storage nodes race to opposite rails, and the winner is decided by the cell's own threshold-voltage mismatch. Reading the settled state therefore yields a random-but-repeatable fingerprint bit. The

central limitation is reliability: cells whose mismatch is small compared with circuit noise are near-metastable and resolve inconsistently, so a fraction of the response bits flip between reads or power-ups.

This project asks a focused question: can a very small SRAM PUF improve its reliability without heavy error-correction hardware, by combining a Schmitt-trigger-assisted readout with a lightweight unstable-bit detector? We build a conventional 6T SRAM PUF as a baseline and a proposed variant that adds a Schmitt-trigger buffer on the read path and a dual-latch detector that identifies unstable bits so they can be masked. Both designs are implemented as a 4×4 array in a 0.5  $\mu\text{m}$  CMOS process and compared using measurable results.

The contributions of this work are: (i) a compact 6T SRAM PUF baseline and a Schmitt-trigger-assisted proposed cell, both laid out and design-rule clean; (ii) a lightweight dual-latch, XOR-based unstable-bit detector; (iii) a mismatch-driven Monte-Carlo evaluation of uniqueness, uniformity and bit-error rate; and (iv) a measured baseline-versus-proposed comparison of area, power, energy, delay and reliability. An important and honest finding is that the reliability improvement comes from the detector-based masking, while the Schmitt trigger mainly improves read-noise immunity.

## II. RELATED WORK

The reference for this project is the ASCH-PUF of He, Li, Yu and Yang [1], a sub-threshold, inverter-chain CMOS PUF reported in the IEEE Journal of Solid-State Circuits. ASCH-PUF attacks the same instability problem but from a different angle: it detects dark (unstable) bits by deliberately skewing the cell voltage, and it offers two stabilization modes, a static mode that stores an enrollment mask and a dynamic mode that needs no non-volatile memory. Through this masking it reports an effectively zero bit-error rate over a wide temperature and voltage range while discarding roughly a third of the raw bits.

We reuse the concept behind that result, that reliability can be won by detecting and removing unstable bits rather than by hardening every cell, but the circuit here is our own. The bit-generating element is an SRAM cell rather than an inverter chain, the readout uses a Schmitt trigger, and the detector is a simplified dual-latch, XOR comparison instead of voltage-skew dark-bit sensing. The distinction between reused inspiration and original design is kept explicit throughout.

## III. PROPOSED DESIGN

### A. Baseline 6T SRAM PUF Cell

The baseline bit-cell is a conventional 6T SRAM cell: two cross-coupled CMOS inverters holding the complementary

nodes Q and QB, with two NMOS access transistors connecting the nodes to the bit-lines under word-line control. No value is written during fingerprint generation; instead the supply is ramped from zero and the cell is allowed to resolve from its metastable start, so the stored bit is decided by device mismatch. SRAM sizing differs deliberately from logic sizing: the pull-down NMOS is made strong ( $3\ \mu\text{m}$ ) and the pull-up PMOS weak ( $1.5\ \mu\text{m}$ ) so that a read is stable and a write is still possible, giving a cell ratio of two. The access transistors are  $1.5\ \mu\text{m}$  and all channel lengths are  $0.6\ \mu\text{m}$ .

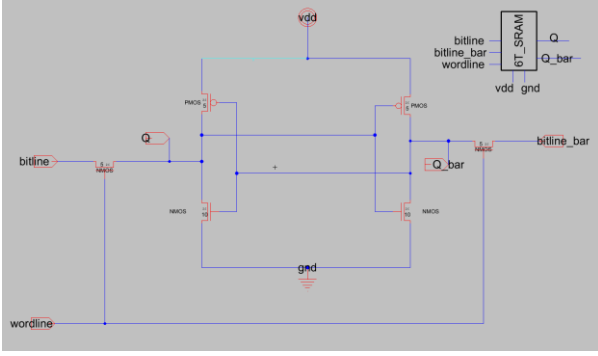


Fig. 1. Baseline 6T SRAM PUF cell.

### B. Baseline 1-bit PUF Cell

Fig. 2 shows the complete baseline 1-bit PUF cell: the resolved storage node is buffered by a plain CMOS inverter readout and captured by an output latch, with no Schmitt trigger or detector. This is the reference against which the proposed cell is measured.

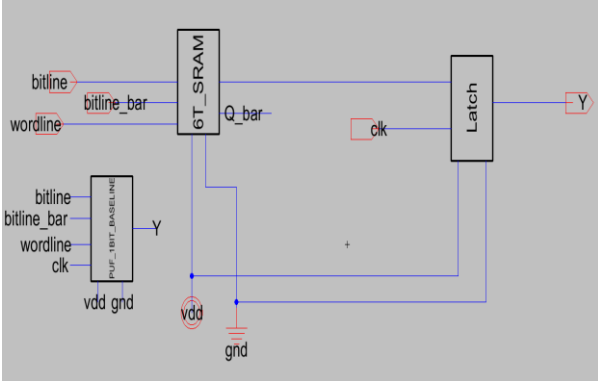


Fig. 2. Baseline 1-bit PUF cell (6T core with inverter readout and latch).

### C. Schmitt-Trigger-Assisted Readout

In the proposed cell the resolved node is read through a six-transistor CMOS Schmitt trigger before the output latch. The Schmitt trigger has two switching thresholds rather than one, so its output only changes after the input has moved well past the mid-rail. This hysteresis rejects small fluctuations on the read path and produces a clean, near-rail-to-rail output. Fig. 4 shows the simulated voltage-transfer characteristic: the two vertical transitions occur near  $1.75\ \text{V}$  and  $3.1\ \text{V}$ , a hysteresis window of about  $1.35\ \text{V}$  that sets the noise immunity of the read.

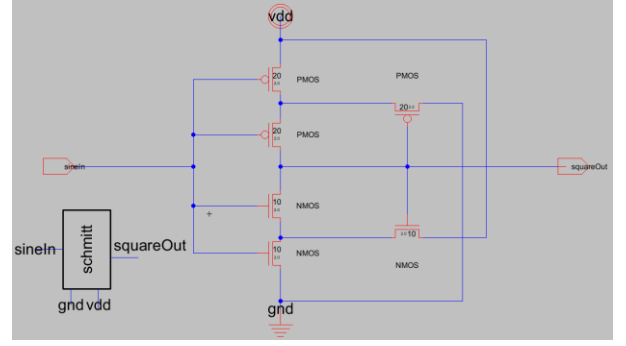


Fig. 3. Six-transistor CMOS Schmitt-trigger readout.

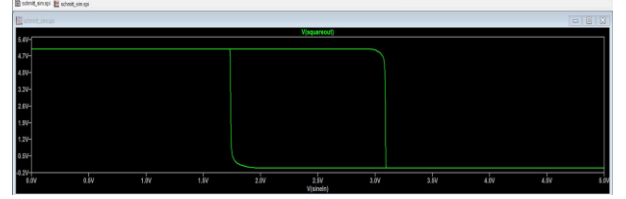


Fig. 4. Simulated Schmitt-trigger voltage-transfer characteristic showing the hysteresis window.

### D. Precharge Circuit

A three-PMOS precharge structure, two devices that charge the bit-line pair and one that equalizes them, brings the bit-lines to a known, balanced state before the word-line is activated. Precharge is a read-preparation step: it makes each read clean, repeatable and full-swing, but it does not create the bit. The fingerprint is generated by the cross-coupled pair settling from its metastable start, which is why all PUF metrics in this work are extracted from the power-up (no-precharge) mode.

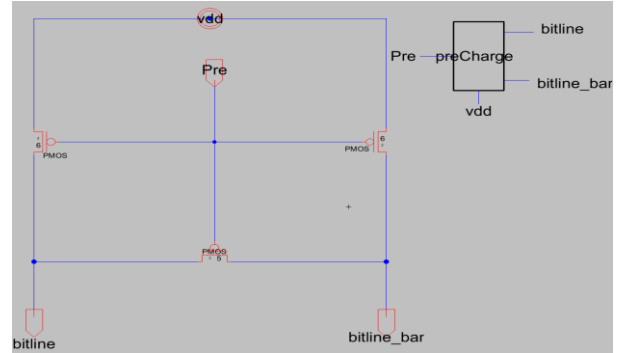


Fig. 5. Three-transistor precharge circuit.

### E. Unstable-Bit Detector

The detector is the element responsible for the reliability improvement. The same cell output is sampled by two latches under slightly different timing and the two samples are compared with an XOR gate. If the readings agree the bit is stable and the XOR output is zero; if they disagree the bit is flagged as unstable. In operation the flagged bits are masked so that only stable bits contribute to the key, mirroring the dark-bit strategy of the reference design with far simpler hardware.

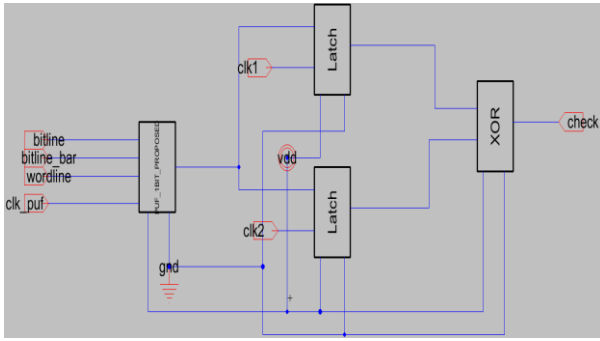


Fig. 6. Dual-latch, XOR-based unstable-bit detector.

### F. Proposed 1-bit PUF Cell

The complete proposed bit-cell assembles the blocks introduced above: the 6T SRAM core generates the raw bit from mismatch, the gated Schmitt trigger buffers the resolved storage node, and the output latch holds the value, while the unstable-bit detector monitors read-to-read stability. Fig. 7 shows the assembled proposed 1-bit cell, the direct counterpart of the baseline cell of Fig. 2.

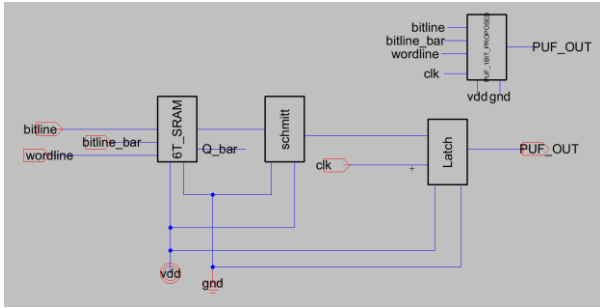


Fig. 7. Proposed 1-bit PUF cell (6T + Schmitt + latch).

### G. 4×4 PUF Array

Sixteen cells are arranged as a 4×4 array with shared word-lines per row and shared bit-line pairs per column, producing a 16-bit response. Row and column selection during simulation is performed directly through the stimulus signals rather than a dedicated decoder or multiplexer; this simplification is noted as a limitation. Fig. 8 and Fig. 9 show the assembled baseline and proposed 4×4 arrays.

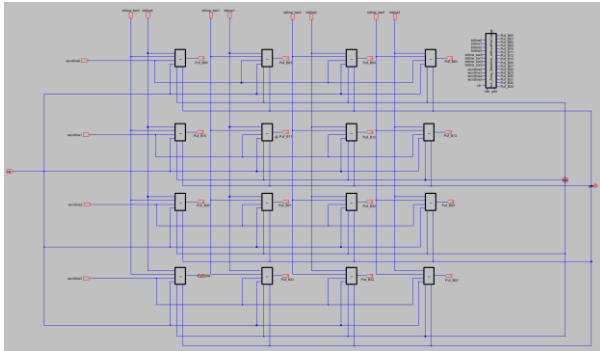


Fig. 8. Baseline 4×4 PUF array.

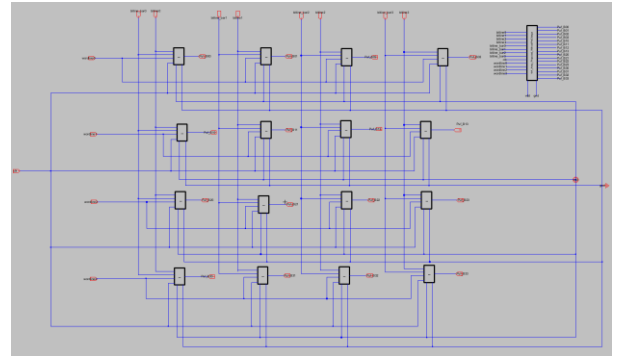


Fig. 9. Proposed 4×4 PUF array.

### H. Transistor Sizing

Table I summarizes the sizing used throughout the design. Logic gates follow the usual wide-PMOS convention for a balanced switching threshold, whereas the SRAM cell uses the read-stability / writability sizing described above. Series-connected devices in cells such as the latch are widened to compensate for stacked on-resistance.

TABLE I Transistor Sizing ( $\lambda = 0.3 \mu\text{m}$ )

| Block           | NMOS W/L ( $\mu\text{m}$ ) | PMOS W/L ( $\mu\text{m}$ ) |
|-----------------|----------------------------|----------------------------|
| CMOS inverter   | 3 / 0.6                    | 6 / 0.6                    |
| 6T pull-down    | 3 / 0.6                    | —                          |
| 6T pull-up      | —                          | 1.5 / 0.6                  |
| 6T access       | 1.5 / 0.6                  | —                          |
| Schmitt trigger | 3 / 0.6                    | 6 / 0.6                    |
| Latch / XOR     | 1.5 / 0.6                  | 3 / 0.6                    |

## IV. SYSTEM ARCHITECTURE

The complete signal flow of the proposed design is: precharge → 6T SRAM cell → Schmitt-trigger readout → output latch → unstable-bit detector (two latches and an XOR). Precharge sets a known start, the cell generates the raw bit from mismatch, the Schmitt trigger cleans the read, the latch holds the value, and the detector decides whether the bit is trustworthy. The baseline design follows the same path without the Schmitt trigger and detector.

## V. SIMULATION SETUP AND EVALUATION METRICS

Schematics and layouts were captured in Electric VLSI and simulated in LTspice XVII using the AMIS C5 0.5  $\mu\text{m}$  BSIM3 models at  $V_{DD} = 5 \text{ V}$ . Fingerprint generation is modeled by a 0-to-5 V supply ramp with the access transistors off, so each cell resolves by its own mismatch. Because the BSIM3 models do not accept the delvto instance parameter in LTspice, threshold-voltage mismatch is injected through a small series gate-offset voltage source on each core transistor, drawn from a Gaussian with  $\sigma = 9 \text{ mV}$  (a Pelgrom-based estimate). A Monte-Carlo sweep re-draws the offsets on every run to emulate independent chips. All reported metrics are schematic-level; only area is taken from the drawn layout.

The response quality and cost figures in the comparison table are computed with the standard PUF metrics below. Equations (1)–(5) are reported as percentages (i.e., multiplied by 100); (6) and (7) give energy and power, and (8) the readout delay.

$$U = \frac{1}{n} \sum_{i=1}^n r_i \quad (1)$$

$$D = \frac{2}{k(k-1)} \sum_{i=1}^{k-1} \sum_{j=i+1}^k \frac{HD(R_i, R_j)}{n} \quad (2)$$

$$BA_l = \frac{1}{k} \sum_{i=1}^k r_{i,l} \quad (3)$$

$$BER = \frac{1}{x} \sum_{t=1}^x \frac{HD(R_{ref}, R_t)}{n} \quad (4)$$

$$Reliability = 100\% - BER \quad (5)$$

$$E = \int V_{DD}(t) \cdot I_{DD}(t) dt \quad (6)$$

$$P_{idle} = V_{DD} \cdot I_{leak} \quad (7)$$

Here  $n$  is the response length in bits,  $k$  the number of chips,  $x$  the number of repeated reads,  $r_i$  the  $i$ -th response bit,  $R_i$  the response word of chip  $i$ , and HD the Hamming distance. Uniformity (1) measures bias toward ones, uniqueness (2) the average inter-chip Hamming distance, and bit-aliasing (3) the per-position bias. Reliability follows from the bit-error rate through (4) and (5); (6) gives the read energy by direct integration of the instantaneous supply power (valid because  $V_{DD}$  ramps 0→5 V), and (7) the idle power from the settled leakage current.

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} \quad (8)$$

The readout delay in Table II is a direct transient measurement rather than a statistical metric. A fast input step (0.1 ns edges) drives the readout element loaded by 10 fF, and (8) averages the high-to-low and low-to-high propagation delays  $t_{pHL}$  and  $t_{pLH}$ , each measured between the input and the output crossing the 2.5 V (50 %) midpoint using LTspice TRIG/TARG. This yields 54 ps for the inverter readout and 101 ps for the Schmitt trigger.

## VI. RESULTS AND DISCUSSION

Functional waveforms confirm correct operation of every block and of the assembled arrays, at both the single-cell and the array level.

### A. Single-Cell Functional Simulation

The 1-bit cells were simulated first. Fig. 10 shows the schematic that drives the baseline and proposed cells together, and Fig. 11 overlays their outputs with the XOR that marks where they differ. Fig. 12 and Fig. 13 give the individual baseline and proposed 1-bit transient responses: both resolve to full rails from the power-up ramp, and the proposed cell delivers a clean, well-defined output through its Schmitt-trigger readout and latch.

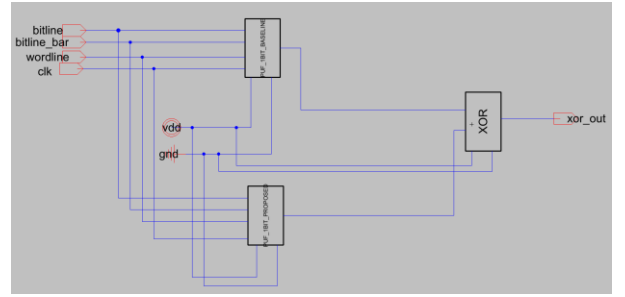


Fig. 10. Baseline-versus-proposed 1-bit comparison schematic.



Fig. 11. Baseline-versus-proposed 1-bit comparison: cell outputs and their XOR.

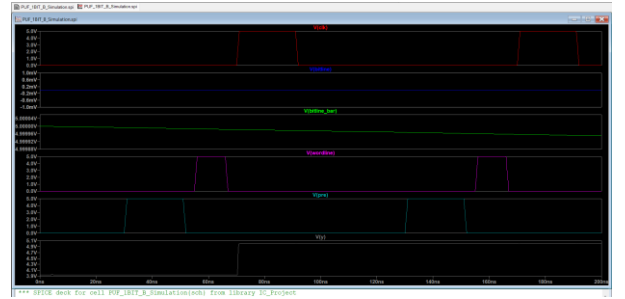


Fig. 12. Transient simulation of the baseline 1-bit cell.

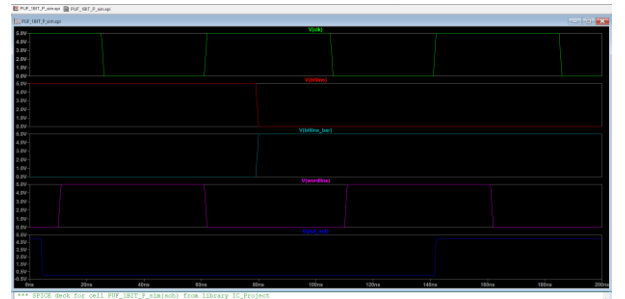


Fig. 13. Transient simulation of the proposed 1-bit cell.

### B. Unstable-Bit Detector Operation

The unstable-bit detector was simulated on its own. Fig. 14 shows its operation: two latches sample the same cell under slightly different timing and the XOR compares the two samples, raising a flag whenever they disagree. This flag is the mechanism used to mask unstable bits and is the source of the reliability improvement reported in Section VII.

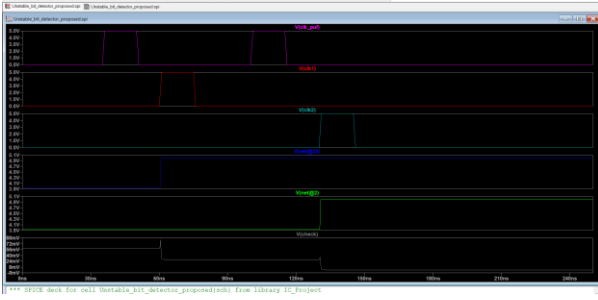


Fig. 14. Unstable-bit detector in operation: dual-latch samples and their XOR flag.

### C. Array Functional Simulation

The same comparison was repeated for the 4×4 array. Fig. 15 and Fig. 16 give the baseline-versus-proposed array comparison schematic and its transient simulation, while Fig. 17 and Fig. 18 show the individual transient responses of the baseline and proposed arrays. In every case all sixteen cells resolve to stable, full-swing values, confirming correct array operation for both designs.

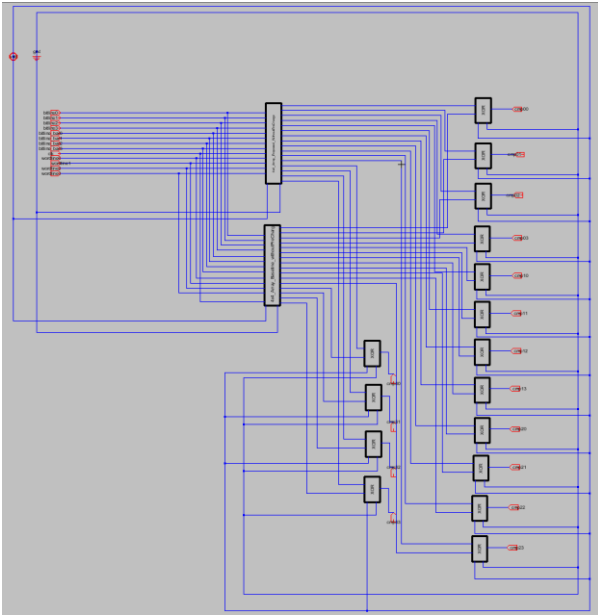


Fig. 15. Baseline-versus-proposed 4×4 array comparison schematic.

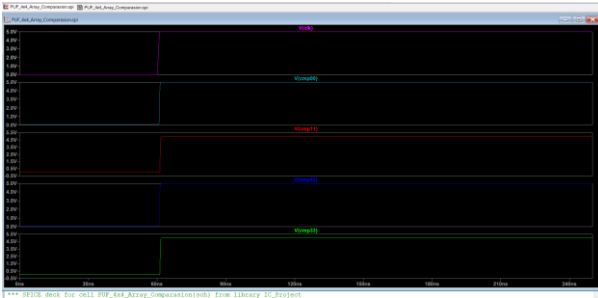


Fig. 16. Baseline-versus-proposed 4×4 array comparison: array outputs and their XOR.

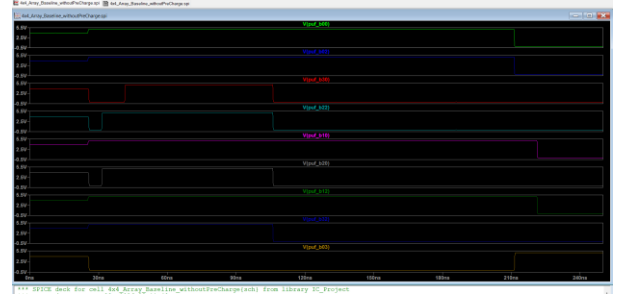


Fig. 17. Transient simulation of the baseline 4×4 array.

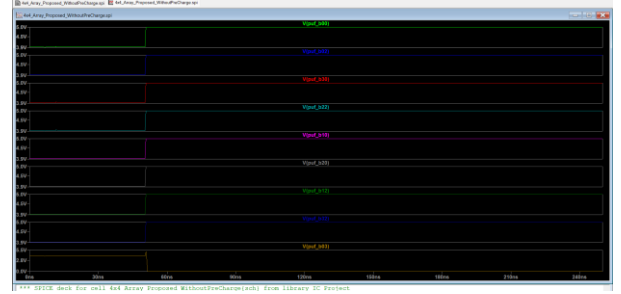


Fig. 18. Transient simulation of the proposed 4×4 array.

### D. Uniqueness and Uniformity

Over 60 Monte-Carlo instances the 4×4 array yields a mean inter-chip Hamming distance (uniqueness) of 49.8 % and a uniformity of 45.9 % ones, with per-position bit-aliasing clustered near 50 %. Fig. 19 shows the resulting uniqueness distribution and the per-position bit-aliasing. Uniqueness is very close to the ideal of 50 % and is highly stable, because it averages 1770 chip pairs. Uniformity is a noisier statistic: across independent 60-chip samples it scatters within a few percent of the ideal 50 % (values of 45.9–50.5 % were observed), and a single-cell Monte-Carlo run over 120 chips gives 47.5 %, consistent with the array. Uniqueness is inherently an array metric because it compares multi-bit words, so it is reported only at the 4×4 level.

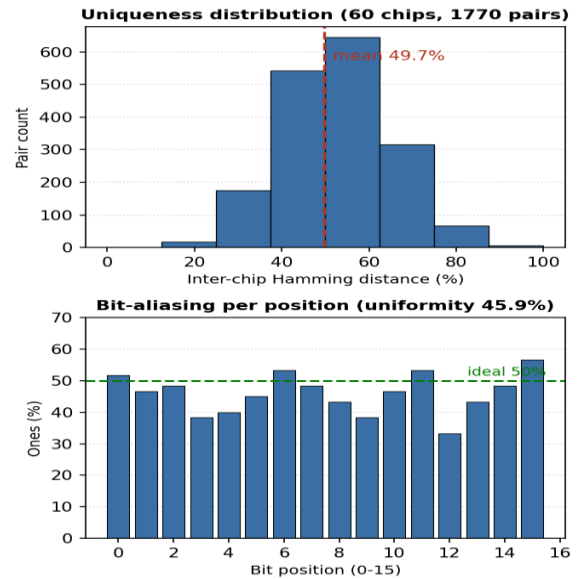


Fig. 19. Monte-Carlo PUF metrics over 60 chips: uniqueness distribution (mean 49.8 %) and per-position bit-aliasing (uniformity 45.9 %).

## VII. RELIABILITY ANALYSIS

Reliability was evaluated by repeated noisy reads. Thirty-two fixed-mismatch cells were read thirty times each with an independent 5 mV Gaussian perturbation added per read, so that near-metastable cells flip while well-decided cells do not. To avoid circularity, the reads were split: the first fifteen reads act as an enrollment set that identifies and masks unstable cells, and the last fifteen reads are used as an independent evaluation set.

On the independent evaluation reads the baseline PUF shows a bit-error rate of 15.0 % (85.0 % reliability). After the detector masks the 59 % of cells that were unstable during enrollment, the retained bits show a BER of 1.5 % (98.5 % reliability), roughly a ten-fold reduction, at the cost of shortening the usable key. The high mask fraction follows directly from the deliberately aggressive noise level relative to the 9 mV mismatch spread; a lower noise level reduces both the BER and the number of masked bits. We note honestly that the Schmitt trigger does not by itself change which way an already-resolved cell settles; its role is to harden the read against noise, whereas the measured reliability gain is produced by the detector-based masking.

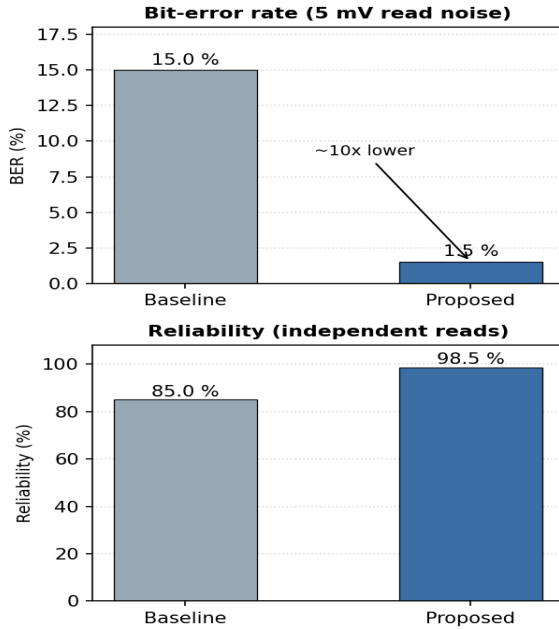


Fig. 20. Bit-error rate and reliability, baseline versus proposed (independent evaluation reads).

## VIII. BASELINE-VERSUS-PROPOSED COMPARISON

Table II collects the measured comparison. Uniqueness and uniformity are properties of the shared 6T core and are therefore equal for both designs. The proposed design pays in area, idle power, read energy and readout delay to obtain the reliability improvement. Power and energy scale as expected by a factor of sixteen from the single cell to the array, which cross-checks the measurements.

TABLE II *Baseline versus Proposed (measured)*

| Metric                     | Baseline                | Proposed                |
|----------------------------|-------------------------|-------------------------|
| Uniqueness (inter-chip HD) | 49.8 %                  | 49.8 %                  |
| Uniformity (% ones)        | 45.9 %                  | 45.9 %                  |
| BER (5 mV read noise)      | 15.0 %                  | 1.5 %                   |
| Reliability                | 85.0 %                  | 98.5 %                  |
| Key bits retained          | 100 %                   | 41 %                    |
| Idle power, 1-bit          | 75.6 pW                 | 216 pW                  |
| Idle power, 4×4            | 1.20 nW                 | 3.43 nW                 |
| Generation energy, 4×4     | 1.98 pJ                 | 1.98 pJ                 |
| Read energy, 4×4 (16-bit)  | ~ 0                     | 105 pJ                  |
| Readout delay              | 54 ps                   | 101 ps                  |
| Cell area, 1-bit           | 5,595 $\mu\text{m}^2$   | 7,339 $\mu\text{m}^2$   |
| Array area, 4×4            | 160,921 $\mu\text{m}^2$ | 244,455 $\mu\text{m}^2$ |

The proposed readout uses a gated Schmitt trigger: header and footer devices disconnect the Schmitt from the rails between reads. Gating is necessary because the classic 6T Schmitt would otherwise draw a large static current while its input sits at a rail; with gating, the idle power of the 1-bit proposed cell is 216 pW instead of milliwatt-level. The remaining read energy of 6.56 pJ per bit (105 pJ for the 16-bit array) is dominated by the Schmitt during the brief active read window, which points to a lower-power readout as future work.

## IX. LAYOUT AND VERIFICATION

Full-custom layouts were drawn for the inverter, 6T cell, Schmitt trigger, latch, XOR, detector, the 1-bit cells and the 4×4 arrays. Fig. 21 shows the 6T cell layout, Fig. 22 and Fig. 23 the Schmitt-trigger and unstable-bit-detector layouts, Fig. 24 and Fig. 25 the baseline and proposed 1-bit cell layouts, and Fig. 26 and Fig. 27 the baseline and proposed 4×4 array layouts. Cell footprints were read directly from the layout: the baseline 1-bit cell is 5,595  $\mu\text{m}^2$  and the proposed 1-bit cell is 7,339  $\mu\text{m}^2$  (+31 %). The measured array footprints are 160,921  $\mu\text{m}^2$  (baseline) and 244,455  $\mu\text{m}^2$  (proposed, +52 %); the larger array-level overhead reflects both the extra per-cell circuitry and the additional inter-cell routing of a hand-drawn, non-compact layout.

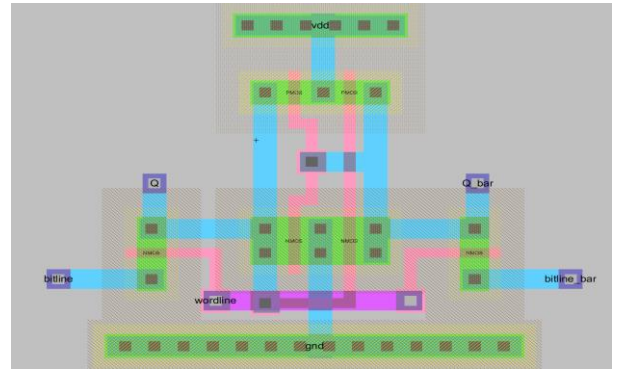


Fig. 21. 6T SRAM cell layout.

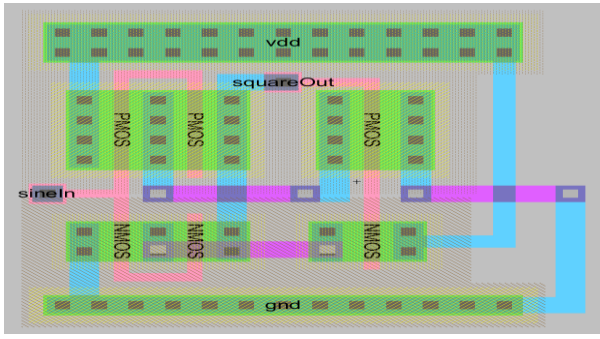


Fig. 22. Schmitt-trigger layout.

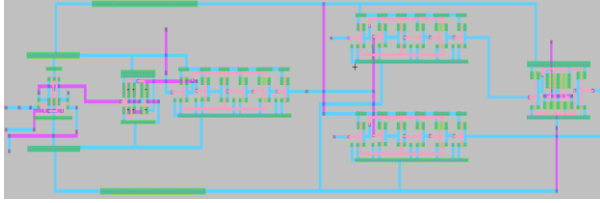


Fig. 23. Unstable-bit detector layout.

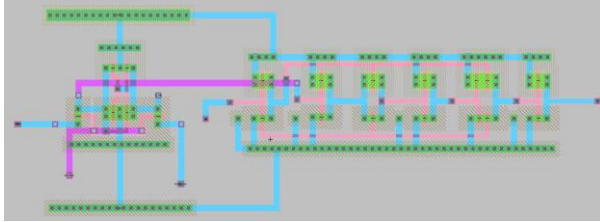


Fig. 24. Baseline 1-bit PUF cell layout.

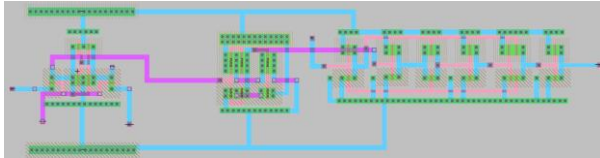


Fig. 25. Proposed 1-bit PUF cell layout.

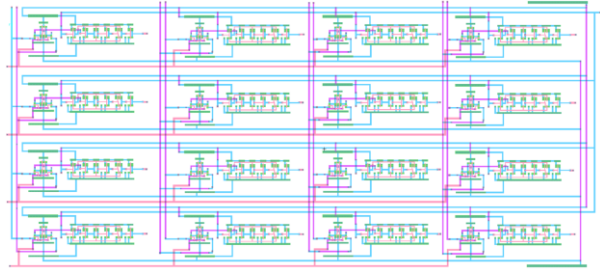


Fig. 26. Baseline 4x4 array layout.

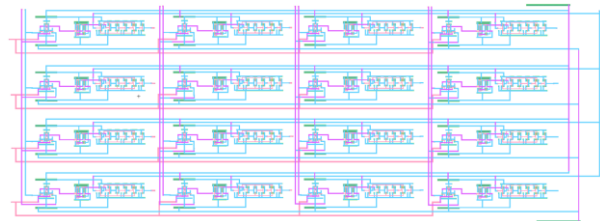


Fig. 27. Proposed 4x4 array layout.

Every implemented cell passed design-rule checking (DRC) and network-consistency checking (NCC) against its schematic, as summarized in Table III.

TABLE III Layout Verification

| Cell                  | DRC   | NCC   |
|-----------------------|-------|-------|
| CMOS inverter         | Clean | Match |
| 6T SRAM               | Clean | Match |
| Schmitt trigger       | Clean | Match |
| Latch                 | Clean | Match |
| XOR                   | Clean | Match |
| Unstable-bit detector | Clean | Match |
| 1-bit PUF cell        | Clean | Match |

## X. CONCLUSION AND FUTURE WORK

A compact 6T SRAM PUF and a Schmitt-trigger-assisted variant were designed, laid out and simulated in a 0.5  $\mu\text{m}$  CMOS process. The array achieves near-ideal uniqueness (49.8 %) and uniformity (45.9 %). A lightweight dual-latch, XOR-based unstable-bit detector lowers the bit-error rate from 15.0 % to 1.5 % on independent reads, about a ten-fold reliability gain, while the Schmitt trigger contributes read-noise immunity and a cleaner output at the cost of roughly double the readout delay. The proposed cell is 31 % larger at the bit level and requires a gated readout to keep idle power low. The evidence supports the central claim: a very small SRAM PUF can be made substantially more reliable through lightweight unstable-bit detection, without heavy error-correction hardware.

The main limitations are that the electrical metrics reported here are taken at schematic level; layout-level simulations of every cell were also carried out and are included in the project files, but are not reproduced in this paper. Reliability was assessed through repeated-read noise rather than full temperature and voltage sweeps, and array selection was driven by stimulus rather than a dedicated decoder. Natural extensions are reporting the full post-layout characterization, temperature and supply sweeps, a compacted and decoder-addressed array, a crowbar-free low-power readout, and formal randomness testing of the response.

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